

ADL - Challenge Report

Avengers Don't Lose

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1 Introduction

In this work, pixel-level anomaly detection in multi-view industrial images is addressed on the **Spacepresso dataset**, released as part of the ADL 2025–2026 anomaly detection challenge. The goal is to accurately identify and segment anomalous regions within multi-perspective image sets across several object categories, including mechanical gears, electronic components, coffee beans, and pistachios, under the Pixel-Level Average Precision (AP) metric.

The dataset presents two compounding challenges: eight different classes with anomalies of very different nature, and an extremely low availability of anomaly ground truths. These cause standard unsupervised methods to struggle, motivating a semi-supervised patch-level approach.

We propose a **semi-supervised patch-MLP ensemble** that extracts dense features from three frozen pretrained backbones and trains a lightweight classifier for each at the patch level. The final ensemble achieves **0.8781** Public AP, outperforming the PatchCore Lite baseline of **0.4040**.

2 Methodology

2.1 Problem Formulation

Given an input image $x \in \mathbb{R}^{H \times W \times C}$, the goal is to predict a dense anomaly score map $\hat{Y} \in$

$[0, 1]^{H \times W}$, where higher values indicate greater anomaly likelihood. Each physical sample is captured from five viewpoints sharing a unique *sample_id*, enabling multi-view fusion at inference time. Since anomalous patches constitute approximately 1% of all training patches, a semi-supervised representation-learning approach is motivated: frozen pretrained backbones supply rich visual priors while lightweight per-class classifiers are trained on the available pixel-level annotations.

2.2 Patch-Level Anomaly Modeling

Three pretrained vision backbones (Table 1) are used to extract dense patch-level features from input images; all are kept frozen during training to preserve the general-purpose visual representation.

The spatial feature map is divided into patches, each represented as a feature vector $p_i \in \mathbb{R}^d$. A lightweight MLP classifier g_θ is trained independently per product class and per backbone to assign each patch an anomaly score $s_i = g_\theta(p_i) \in [0, 1]$. Class-wise training is necessary because defect appearances differ substantially across categories. Patch targets are derived by aggregating ground-truth pixel mask labels within each patch region, with masks resized using nearest-neighbor interpolation to preserve binary boundaries.

To address the extreme class imbalance, training minimizes Focal Loss ($\gamma = 2$, α auto-computed from

Table 1: Pretrained backbones used in the ensemble. All images are resized to 224×224 ; feature dim refers to the concatenated intermediate feature vector per patch.

Backbone	Type	Grid	Feat. dim	Key strength
WideResNet50-2 [6]	CNN (ImageNet)	28×28	1536	Precise local texture anomalies
DINOv2 ViT-B/14 [3]	ViT (self-supervised)	16×16	3072	Semantic generalization to unseen defect types
ConvNeXT-Large [2]	CNN (ImageNet)	28×28	1152	Complementary CNN inductive bias to WideResNet50-2

the class ratio), which focuses learning on the rare anomalous patches critical for precise localization.

Prior to training, patch features are **z-score normalized** using the mean and standard deviation computed over normal patches only, ensuring a stable input distribution for the MLP regardless of backbone scale.

To further sharpen the decision boundary, DINOv2-based classifiers additionally perform a **Hard Negative Mining (HNM)** pass after initial training: a second iteration identifies normal patches scored most confidently as anomalous and reintroduces them with corrected labels, effectively reducing false positives in ambiguous regions.

2.3 Ensemble and Multi-View Fusion

The three backbone classifiers are combined via **per-pixel max fusion**: for each test image, the anomaly maps from WR50, DINOv2, and ConvNeXT-Large are merged by taking the element-wise maximum. Max fusion is preferred over averaging because a defect detected confidently by any single backbone contributes its full score to the final map without being diluted by low responses elsewhere.

Multi-view fusion is then applied at the sample level: all five views are scaled up to match the peak score of the strongest-scoring view, ensuring anomalies visible from at least one viewpoint are reflected across the full multi-view set.

2.4 Training and Inference Setup

During training, a **per-class background suppression** mask zeros out pixels outside of the object region, reducing false positives in non-informative border areas. WideResNet50-2 and ConvNeXT-Large apply rotational **augmentation** (0° , 90° ,

180° , 270°) combined with flips for rotationally symmetric classes to multiply the limited anomaly training set. Instead, DINOv2 relies exclusively on **Hard Negative Mining**: HNM is not applied to the other two backbones as augmented anomaly patches introduced in the first pass would be misidentified as hard negatives in the second, degrading performance.

At test time, patch-level predictions are upsampled to image resolution and smoothed with a Gaussian filter. **Test-time augmentation (TTA)** over D4 rotations and flips is applied, with predictions averaged across augmented views to reduce directional bias. Mean aggregation outperformed max significantly on the validation set.

3 Method Development and Results

PatchCore Lite [4] serves as the primary unsupervised baseline (**0.4040**). Ensembling multiple PatchCore models via k-fold aggregation improved only marginally (**0.4286**), confirming the ceiling of unsupervised feature matching under limited normal data. EfficientAD [1] and DRAEM [7] were evaluated locally but discarded due to poor validation performance. Switching to a semi-supervised patch MLP trained directly on ground-truth masks yielded an immediate gain to **0.6900**. Rotational and flip augmentations on the scarce anomaly images, combined with cosine LR decay, pushed this to **0.7949**. TTA over D4 rotations and flips then raised the score to **0.8245**.

Ensemble strategies were explored next. Replacing the single WR50 model with a WR50 + FastFlow [5] combination yielded negligible gain on the public leaderboard (0.8252), despite stronger local validation scores, suggesting overfitting to the

Table 2: Public leaderboard results on the Spacepresso dataset. WR50 denotes WideResNet50-2.

Method	Backbone / Configuration	Public AP
Semi-supervised Patch MLP	WR50	0.6900
+ rotation & flip aug., LR decay	WR50	0.7949
+ TTA	WR50	0.8245
Ensemble	WR50 + DINOv2 ViT-S	0.8559
Ensemble	WR50 + DINOv2 ViT-B	0.8674
Ensemble (final)	WR50 + DINOv2 ViT-B + ConvNeXT-L	0.8781

held-out distribution. Replacing FastFlow with a DINOv2 ViT-S semi-supervised MLP produced a much larger jump to **0.8559**, confirming that a self-supervised ViT backbone contributes complementary features to CNN-based extractors. Upgrading to ViT-B improved further to **0.8674**, and adding ConvNeXT-Large as a third backbone yielded the final score of **0.8781**. All results are in Table 2.

4 Discussion

The strong performance of the proposed approach reflects its alignment with the specific challenges of Spacepresso. Frozen pretrained backbones transfer well to unseen defect types without requiring large anomalous training sets, directly addressing the scarcity of labeled anomalies. At the ensemble level, CNN and transformer backbones proved complementary: WR50 and ConvNeXT-L captured local texture deviations while DINOv2 provided semantic consistency and robustness to structurally novel defect types. Figure 1 illustrates the typical ensemble behaviour.

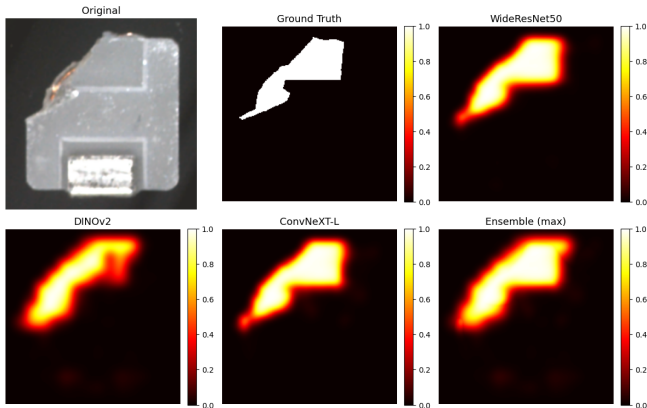


Figure 1: Anomaly localization example. Colorbars indicate normalized anomaly scores in $[0, 1]$.

A notable case is that of classes 07 and 08, where anomalies span the entire object shape: DINOv2 consistently fails to detect these while the CNN backbones do, making the ensemble critical for maintaining recall in those classes. Max fusion preserves localized high-confidence detections from any single backbone without dilution, but propagates false positives: any spurious detection from a single backbone appears in the final map.

The method assumes anomalies manifest as local patch-level deviations and that multi-view consistency is informative; these assumptions hold on Spacepresso but may not generalize to all industrial settings. Inference cost is also high due to three large backbones combined with TTA and multi-view processing.

5 Conclusions

A semi-supervised patch-MLP ensemble is proposed, directly leveraging pixel-level anomaly annotations to train lightweight per-class classifiers on top of three complementary frozen backbones, with Hard Negative Mining applied exclusively to DINOv2 in place of augmentation, achieving **0.8781** Public AP on the Spacepresso dataset, more than $2\times$ the PatchCore Lite baseline of **0.4040**.

Future work could explore fusion strategies between max and average that balance false positive suppression with true detection preservation; improve multi-view aggregation by learning which viewpoints are most informative per class rather than treating all five views equally; adopt class-adaptive backbone weighting motivated by DINOv2’s observed failure on whole-shape anomalies in classes 07–08; and leverage the provided textual anomaly descriptions through language-guided methods such as CLIP-based conditioning to improve generalization to under-specified defect types.

References

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